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Investigation of molten pool flow behavior and porosity formation mechanisms in laser welding under different oscillation modes

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Highlights

- A 3D transient thermo-fluid-interface model reveals melt flow and porosity mechanisms.
- Laser oscillation homogenizes heat input and reduces surface temperature gradients.
- Vertical oscillation weakens upward vortices and stabilizes keyhole walls.
- Circular oscillation forms circumferential flow and minimizes keyhole depth fluctuation.

- High-frequency oscillation enables bubble recapture and effective porosity suppression.

Abstract

This study systematically analyses the melt pool flow behaviour and porosity formation mechanisms during laser welding of aluminium alloy plates. A three-dimensional transient numerical model coupling thermal, fluid flow, and interface phenomena was established to investigate the melt pool flow characteristics, keyhole dynamic evolution, and porosity generation processes under different laser oscillation modes. A composite heat source model, consisting of a Gaussian surface heat source and a rotating Gaussian volumetric heat source, was employed to accurately characterise the laser energy deposition under non-oscillating, vertical oscillating, and circular oscillating conditions. Numerical results indicate that laser oscillation effectively homogenizes the heat input distribution within the molten pool, significantly reduces surface temperature gradients, and enhances flow stability. Compared to non-oscillating welding, vertical oscillation attenuates the upward recirculation flow within the molten pool, mitigating localised deformation of the keyhole wall. Circumferential oscillation induces stable circumferential flow patterns, significantly suppressing keyhole contraction and collapse, thereby effectively reducing fluctuations in keyhole depth. Under high-amplitude, high-frequency oscillation conditions, the periodic motion of the keyhole promotes the dynamic entrapment and escape of bubbles, substantially reducing the likelihood of porosity and inclusion formation during solidification. This study elucidates the intrinsic mechanism of porosity suppression in oscillating laser welding from a fluid dynamics perspective, providing a theoretical basis for optimising oscillation parameters in aluminium alloy laser welding.

Keywords

CRediT authorship contribution statement

Declaration of competing interest

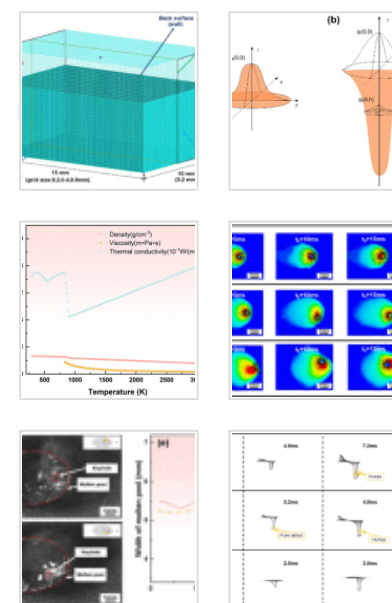
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Oscillating laser welding; Three-dimensional transient numerical simulation; Molten pool fluid dynamics; Keyhole dynamics; Porosity inhibition mechanism

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1. Introduction

Aluminium alloys are widely used in numerous engineering fields due to a favorable combination of properties. Their low density and high strength contribute significantly to weight reduction in structural designs. Furthermore, characteristics such as excellent corrosion resistance, good machinability, and high thermal conductivity make them suitable for applications in the automotive and aerospace industries, where energy efficiency and emission reduction are critical priorities [\[1\]](#), [\[2\]](#), [\[3\]](#). In the fabrication of structural components based on aluminum alloys, welding constitutes a critical process governing joint integrity and in-service reliability. Compared with conventional arc welding, laser welding offers distinct advantages for joining aluminum alloys, characterized by high energy density, fast processing rates, and a narrowly confined heat-affected zone, and minimal workpiece distortion, positioning it as a promising technique for efficient, high-quality manufacturing [\[4,5\]](#). Nevertheless, the high reflectivity and thermal conductivity of aluminum alloys, combined with the complex thermophysical interactions during welding, present notable challenges. Laser welding of these alloys is consequently susceptible to defects such as irregular bead geometry, undercut, porosity, solidification cracking, and joint discontinuities. These imperfections can substantially compromise the mechanical properties and long-term structural integrity of the welded joints [\[6\]](#), [\[7\]](#), [\[8\]](#).

Laser welding of aluminum alloys has been thoroughly explored from perspectives of energy distribution, molten pool behaviors, and laser oscillation strategies. Zhang et al. [\[9\]](#) reported that uneven distribution of energy density can lead to significant asymmetry

in the morphology of the molten pool, causing periodic fluctuations in surface tension. Consequently, the solidification of molten metal is disturbed, ultimately generating defects such as undercuts. Laser beam oscillation has attracted great attention as it can improve weld morphology. Nevertheless, inappropriate oscillation parameters may exacerbate the heterogeneity of energy distribution, resulting in increased density of undercuts [10]. Chuang et al. [11] investigated the welding of magnesium alloys and found that the introduction of an oscillating laser led to an increase of joint elongation by 145.3%, and its stirring effect on the molten pool promoted grain refinement, thereby significantly enhancing joint strength and ductility. Furthermore, Wu et al. [12] reported that circularly oscillating laser welding can greatly reshape the energy density distribution, resulting in even energy distribution within the target zone. The peak power density under circular oscillation laser welding was approximately 13% of that under traditional laser welding. The effective reduction of peak energy suppressed the tendency for welds to become excessively deep or narrow and favored the improvement of overall forming quality. Additionally, Fetzer et al. [13] investigated longitudinal, transverse sinusoidal, and circular oscillations and found that circular oscillation is the most effective pattern in terms of porosity reduction. This can be attributed to the fact that it can enhance the capacity of keyholes to reabsorb bubbles, thereby effectively improving the efficiency of pore elimination.

As experimental approaches can hardly capture the high-speed dynamic processes in laser welding, numerical simulation has been employed to clarify its mechanism. Numerical simulations not only overcome the limited spatial and temporal resolution of experimental observations but also allow access to variables that are difficult to measure directly (e.g., temperature field, velocity field, and pressure field), thereby providing a reliable pathway to explore keyhole dynamics, heat transfer, and flow behaviors of the molten pool. Xia et al. [14] reported a dual-laser beam oscillation strategy for synchronized bilateral welding and developed a CFD model for the investigation of molten pool fluid dynamics and keyhole evolution. This method can significantly optimize energy density distribution, reduce the peak temperature in the T-joint molten

pool, and mitigate the disturbance of metal flow on the keyhole, thereby effectively suppressing bubble formation and reducing porosity. Cho et al. [15] further demonstrated that fully penetrated keyholes are exposed to transient collapses, and their collapse behaviors are closely related to oscillation frequency components. As a result, the frequency characteristics of molten pool instability were clarified. Meanwhile, Deng et al. [16] reported that when laser absorption is insufficient, the molten metal is exposed to low fluidity and uneven thermal distribution. As a result, the keyholes easily collapse, dynamic fluctuations of the molten pool are intensified, and the probability of pore formation increases, ultimately manifesting as exacerbated structural instability during welding.

This study systematically investigated the effects of laser oscillation modes on molten pool dynamics and pore formation during laser welding. The objective was to clarify how different oscillation patterns influence energy distribution, melt flow behavior, and keyhole stability, thereby governing porosity evolution. By varying oscillation amplitude and frequency, their impacts on molten pool stability and pore suppression were quantitatively evaluated. The results provide mechanistic insight and practical guidance for oscillation parameter optimization in laser welding.

2. Numerical simulation

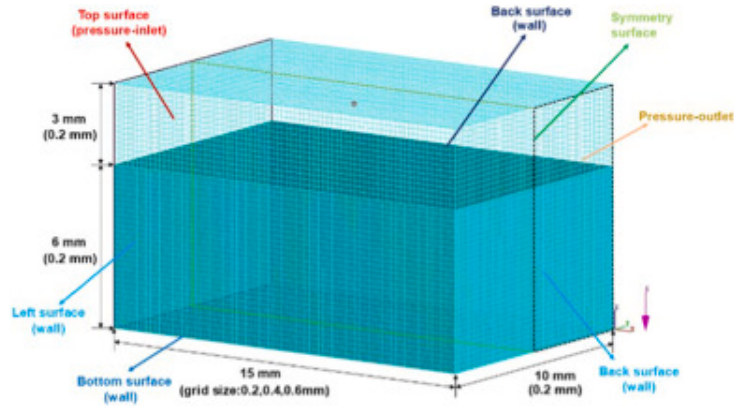
2.1. Grid and hypotheses

The transient thermo-fluid-interface evolution during laser welding is highly complex, involving laser energy absorption and loss, material phase transitions, continuous deformation of the keyhole surface, and strongly coupled flow characteristics within the molten pool. It should be noted that the following assumptions are primarily applicable to partial-penetration (non-full-penetration) laser welding conditions. To improve numerical efficiency while ensuring computational accuracy, appropriate physical simplifications are adopted in this study, with the following assumptions.

- (1) Molten metal is assumed to be an incompressible Newtonian fluid, with the flow in the molten pool mainly laminar.
- (2) The keyhole and molten pool dynamics are primarily governed by surface tension and vapor recoil pressure, with other secondary mechanical effects being neglected.
- (3) The process of welding is approximated as pseudo-steady-state, with the early keyhole formation stage omitted from the simulation.
- (4) The material is isotropic, with thermophysical properties depending solely on temperature, and the coupled effects of pressure on thermal and rheological behavior are not considered.
- (5) Under high-frequency laser oscillation, the keyhole geometry evolves rapidly and exhibits strongly transient behavior. In this case, the optical path inside the keyhole hardly reaches a quasi-steady state, and the cumulative effect of multiple reflections is weakened. Previous studies have shown that, even without explicit ray-tracing algorithms, equivalent volumetric or modified Gaussian heat source models can reasonably reproduce melt pool morphology and thermo-fluid behavior [17]. Therefore, to balance physical fidelity and computational efficiency, multiple internal reflections are neglected, and the laser energy input is modeled as a rotating volumetric Gaussian heat source.

Based on the above assumptions, a three-dimensional computational domain for oscillating laser welding is established, with its mesh structure shown in Fig. 1. Welding is initiated at a position 4 mm away from the origin in the x-direction. The domain is discretized using a non-uniform structured grid: the mesh size in the heat-affected zone is set to 0.2 mm, while the mesh size in the fusion zone and outer regions varies between 0.2 mm and 0.6 mm. The size ratio between adjacent cells is strictly controlled to be less

than 1.4 to ensure accurate numerical capture of temperature gradients and interface deformation. The final model consists of 122,850 cells, providing sufficient resolution while meeting the large time-scale requirements for oscillating laser welding simulations.



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Fig. 1. Numerical simulation of oscillating laser welding: three-dimensional mesh generation.

2.2. Fluid control equation

A 3D numerical model for the oscillating laser welding process is established within a Cartesian coordinate system. The equations governing fluid flow, derived from mass, momentum, and energy conservation principles, are presented below [18].

The continuity equation:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0 \quad (1)$$

The momentum conservation equations:

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u u)}{\partial x} + \frac{\partial(\rho u v)}{\partial y} + \frac{\partial(\rho u w)}{\partial z} = -\frac{\partial P}{\partial x} + \frac{\partial}{\partial x} \left(\mu \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial z} \left(\mu \frac{\partial u}{\partial z} \right) \quad (2)$$

(3)

$$\frac{\partial(\rho v)}{\partial t} + \frac{\partial(\rho v u)}{\partial x} + \frac{\partial(\rho v v)}{\partial y} + \frac{\partial(\rho v w)}{\partial z} = -\frac{\partial P}{\partial y} + \frac{\partial}{\partial x} \left(\mu \frac{\partial v}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial v}{\partial y} \right) + \frac{\partial}{\partial z} \left(\mu \frac{\partial v}{\partial z} \right) + S_v$$

(4)

$$\frac{\partial(\rho w)}{\partial t} + \frac{\partial(\rho w u)}{\partial x} + \frac{\partial(\rho w v)}{\partial y} + \frac{\partial(\rho w w)}{\partial z} = -\frac{\partial P}{\partial z} + \frac{\partial}{\partial x} \left(\mu \frac{\partial w}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial w}{\partial y} \right) + \frac{\partial}{\partial z} \left(\mu \frac{\partial w}{\partial z} \right) + S_w$$

The conservation of energy equation:

(5)

$$\frac{\partial(\rho H)}{\partial t} + \frac{\partial(\rho u H)}{\partial x} + \frac{\partial(\rho v H)}{\partial y} + \frac{\partial(\rho w H)}{\partial z} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + S_E$$

Where ρ is the material density; t is time; u, v, w are velocity components in the x, y, z directions; P is pressure; μ denotes dynamic viscosity; S_u, S_v, S_w are momentum source terms in the x, y, z ; H represents mixture enthalpy; T is the fluid temperature; k stands for thermal conductivity; and S_E indicates the energy source term.

2.3. Fluid free surface tracking

The Volume-of-Fluid (VOF) method, widely employed in multiphase flow simulations, is adopted in this study to track the free interface of the keyhole. This method is based on the concept of determining the gas-liquid interface location by solving for the volume fraction $F(x, y, z, t)$ within each computational cell. The cell is completely occupied by the gas phase when $F=0$, and entirely by the liquid phase when $F=1$. Cells where $0 < F < 1$ contain the free interface between the two phases. This behavior is mathematically expressed as follows [19]:

(6)

$$\frac{dF}{dt} = \frac{\partial F}{\partial t} + u \frac{\partial F}{\partial x} + v \frac{\partial F}{\partial y} + w \frac{\partial F}{\partial z}$$

Where u, v , and w are the velocity components in the x, y , and z directions.

In welding simulations, an accurate description of the solid-liquid interface behavior is essential alongside the treatment of the gas-liquid interface. As liquid metal exhibits a non-zero flow velocity while the solid region remains stationary, significant momentum

damping occurs within the solidification zone. To model this process, this work employs the enthalpy–porosity technique to describe molten pool solidification. The phase change region is modeled as a porous medium with uniform permeability in this approach, where the porosity is directly equated to the liquid volume fraction, varying between 0 and 1. By introducing an additional term incorporated into the momentum equations, the flow resistance at the solid-liquid interface is effectively represented, enabling a physically consistent description of the solidification process and interface motion.

2.4. Initial conditions and boundary specifications

Before the solution of the governing equations, the relevant variables are assigned initial conditions. At the start of the simulation, the entire workpiece is initialized at the ambient temperature, and the fluid velocity throughout the computational domain is set to zero. This is mathematically expressed as follows:

$$T(x, y, z, 0) = T_{ref} \quad (7)$$

$$u_0 = v_0 = w_0 = 0 \quad (8)$$

Where T_{ref} denotes the ambient (reference) temperature, which is set to 300 K in this work.

Moreover, numerical simulations in this study are conducted within a Cartesian coordinate system. The laser heat source is initially positioned at the coordinate origin, with welding proceeding along the positive X-axis direction. To account for the role of gravity on the dynamics of molten pool flow, the gravitational acceleration is set to 9.8 m/s^2 acting in the negative Z-direction. For simplicity in handling the free surface, minor fluctuations on the upper surface of the molten pool are neglected, and the region outside the keyhole is treated as a flat horizontal plane. The corresponding velocity boundary conditions are defined as follows:

$$\mu \frac{\partial u}{\partial z} = -\frac{\partial \gamma}{\partial T} \frac{\partial T}{\partial x}, \mu \frac{\partial v}{\partial z} = -\frac{\partial \gamma}{\partial T} \frac{\partial T}{\partial y}, w = 0 \quad (9)$$

Where u , v , and w are the velocity components in the x , y , and z directions, and $\frac{\partial \gamma}{\partial T}$ is the temperature gradient of surface tension. The outward flow velocity at the molten pool surface may be disregarded; hence, the velocity component w may be set to zero.

The thermal boundary condition [20]:

$$k \frac{\partial T}{\partial \vec{n}} = q - h_c (T - T_{ref}) - \varepsilon \sigma (T^4 - T_{ref}^4) - q_{evp} \quad (10)$$

Where q denotes the laser heat flux, $\vec{n}$ represents the normal vector, h_c represents the convective heat transfer coefficient, ε is the emissivity, σ denotes the Stefan–Boltzmann constant, and q_{evp} corresponds to the heat loss caused by evaporation.

The welds simulated in this study are all non-penetration welds. Consequently, for the lower surface and side boundaries, velocity is absent and evaporative heat transfer need not be considered. Only the thermal effects arising from convection and radiation require consideration, expressed as follows [20]:

$$k \frac{\partial T}{\partial \vec{n}} = -h_c (T - T_{ref}) - \varepsilon \sigma (T^4 - T_{ref}^4) \quad (11)$$

2.5. Source term equation

In the momentum conservation equations, the source term incorporates the contributions of various driving forces acting within the molten pool. Specifically, the pronounced temperature gradients induce spatial density variations of the molten metal as a result of density variations with temperature, giving rise to a thermal buoyancy force. To represent this effect, the classical Boussinesq approximation is adopted in this work to account for the influence of temperature variation on the buoyancy term, which is expressed mathematically as follows [21]:

$$F_b = \rho g \beta (T - T_{ref}) \quad (12)$$

Where g represents the gravitational acceleration, and β denotes the coefficient of thermal expansion for the metal.

Furthermore, it also considers the influence of back pressure and surface tension. The formula for calculating the back pressure caused by melt evaporation is as follows [22]:

$$P_r = A_0 B_0 T_s^{-\frac{1}{2}} \exp\left(-\frac{M_a L_v}{N_a k_a T_s}\right) \quad (13)$$

Where A_0 is a computational compensation coefficient related to ambient pressure, taken as 0.55 in this paper; B_0 is the evaporation coefficient, whose value is material-dependent; T_s denotes the keyhole wall surface temperature; M_a denotes atomic mass; L_v is the latent heat of evaporation; N_a signifies Avogadro's constant; and k_a is Boltzmann's constant.

The temperature dependence of surface tension is described by the following equation [23]:

$$P_\sigma = K\delta(T) \quad (14)$$

$$\delta(T) = \delta_0 + A_\delta(T - T_m) \quad (15)$$

Where K denotes the curvature of the free surface; δ_0 represents the surface tension coefficient; A_δ represents the temperature gradient of surface tension; and T_m denotes the liquidus point of the material.

Combined with momentum dissipation at the solid–liquid boundary, both the enthalpy pore model and the rebound pressure and buoyancy are incorporated into the momentum source term for consideration. The mathematical expression is as follows [24]:

$$S_u = -\frac{\mu}{J}(u - u_0) + P_{rx} + P_{\sigma x} \quad (16)$$

$$S_v = -\frac{\mu}{J}v + P_{ry} + P_{\sigma y} \quad (17)$$

$$S_w = -\frac{\mu}{J}w + P_{rz} + P_{\sigma z} + \rho g \beta(T - T_{ref}) \quad (18)$$

Where g denotes gravitational acceleration, and J represents the permeability of the

porous medium for the 5A06 aluminium alloy, which value can be obtained using the Kozeny-Carman equation:

$$J = \frac{f_l^3 + \delta}{A(1 - f_l^2)} \quad (19)$$

Where A is the coefficient representing the liquid fraction; f_l denotes the liquid phase volume fraction in the mixing zone, which may be regarded as a function of temperature.

For the mixed enthalpy H in the energy conservation equation, the enthalpy of fusion during the welding process must be taken into account. Its mathematical expression may be represented as follows [25]:

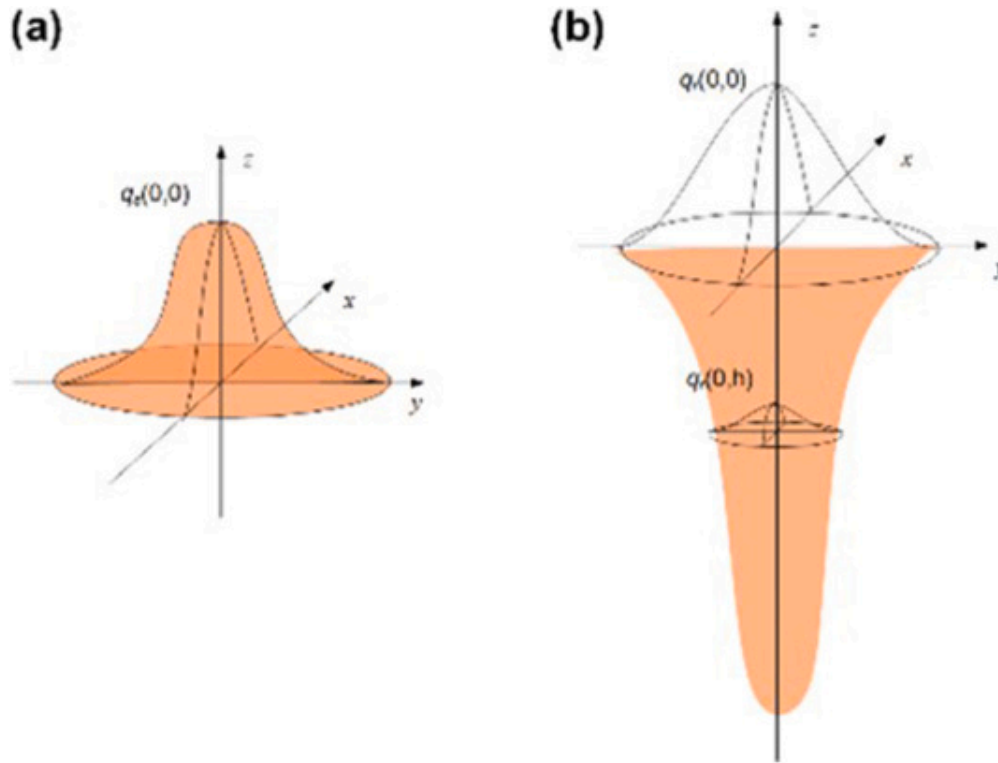
$$H = + \int_{T_{ref}}^T C_p dT + f_l L_m \quad (20)$$

Where h_{ref} denotes the reference enthalpy, C_p represents the specific heat of the material, and L_m denotes the fusion latent heat.

The energy source term S_E may be regarded as the thermal influence of laser energy on the material during heat transfer, along with energy losses incurred throughout the process. Its expression may be described as follows [26]:

$$S_E = q(x, y, z) - \frac{\partial}{\partial t} (\rho \Delta H) - \frac{\partial}{\partial x} (\rho u \Delta H) - \frac{\partial}{\partial y} (\rho v \Delta H) - \frac{\partial}{\partial z} (\rho w \Delta H) \quad (21)$$

In this work, the transfer of energy from laser irradiation was modeled using a combined approach. A Gaussian surface heat source was applied to represent the energy deposition through the plume, and a volumetric heat source with a rotating Gaussian distribution was employed to simulate the Fresnel absorption of the laser energy inside the material. The spatial distribution of the energy input within a cartesian coordinate system is presented in Fig. 2.



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Fig. 2. Schematic of the oscillating laser heat source: (a) Cylindrical heat source of plasma; (b) Rotary-Gaussian body heat source of laser beam.

To accommodate the distinct laser energy attenuation behaviors under different oscillation modes, energy distribution coefficients α_1 and α_2 are introduced to quantify the fraction of total energy assigned to the Gaussian surface heat source and the rotating Gaussian volumetric heat source, respectively, with $\alpha_1 + \alpha_2 = 1$. Additionally, correction factors η_1 and η_2 are integrated into the heat source formulation to improve the model's capability of replicating actual weld bead geometry. These factors adjust the energy distribution to achieve closer agreement with the experimentally observed cross-sectional profile of the weld.

The mathematical expression for the oscillating laser composite heat source is given by the following equation [27]:

$$q_s(x, y, z) = \frac{\alpha_1 Q}{\pi r_s^2} \exp\left(-\frac{3\eta_1(x^2 + y^2)}{r_s^2}\right) \quad (22)$$

$$q_r(x, y, z) = \frac{9\alpha_2 Q}{\pi r_r^2 h_r} \frac{e^3}{e^3 - 1} \exp\left(-\frac{9\eta_2(x^2 + y^2)}{r_r^2 \ln(h_r/z)}\right) \quad (23)$$

$$q_{total}(x, y, z) = q_v(x, y, z) + q_r(x, y, z) \quad (24)$$

Where $q_{total}(x, y, z)$ denotes the total laser energy input to the material; $q_s(x, y, z)$ and $q_r(x, y, z)$ represent the energy distribution expressions for the columnar heat source and rotating Gaussian heat source respectively; Q denotes the laser power; r_s is the effective radius of the Gaussian surface heat source, set to twice the spot diameter (0.6 mm); r_r approximates the effective radius of the rotating Gaussian heat source to the spot radius (0.3 mm); h_c denotes the effective height of the columnar heat source, equivalent to the plume obstruction height previously quantified, while h_r represents the effective depth of the rotating Gaussian heat source. Correction factors η_1 and η_2 are employed to refine the model outcomes until achieving an accuracy exceeding 92%. Table 1 details the heat source parameter configurations corresponding to various oscillation modes.

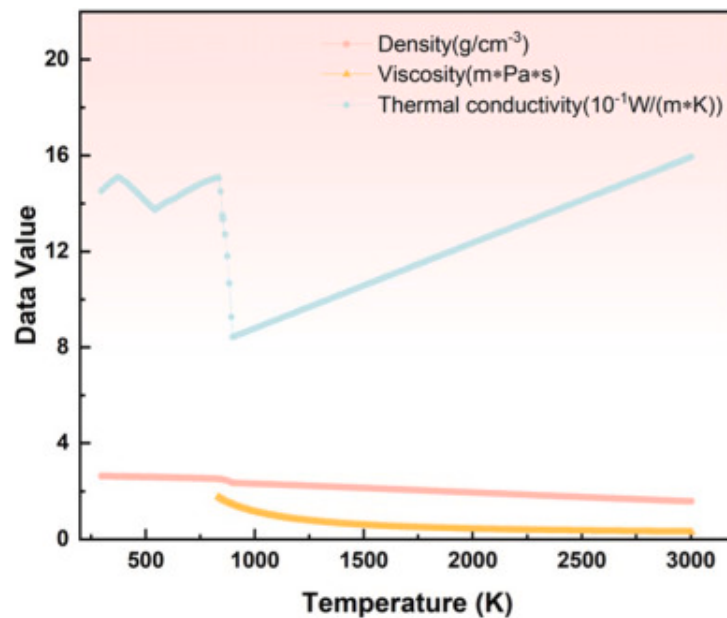
Table 1. Heat source parameter settings corresponding to different oscillation modes.

Oscillation modes	Non-oscillation	Vertical oscillation	Circular oscillation
α_1	0.13	0.07	0.03
α_2	0.87	0.93	0.97
η_1	1	0.9	0.9
η_2	0.30	0.15	0.15

Oscillation modes	Non-oscillation	Vertical oscillation	Circular oscillation
h_r	4.5 mm	3.5 mm	3.0 mm

2.6. Thermal properties of materials

During welding, key material properties vary with temperature, which consequently alters heat and mass transport in the molten pool. Accordingly, JmatPro was employed to calculate the temperature-dependent thermophysical properties. The software simultaneously calculated the material's density, thermal conductivity, specific heat capacity, and viscosity, as shown in Fig. 3. Remaining material properties were treated as temperature-independent constants, summarized in Table 2 [5].



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Fig. 3. Thermo-physical properties of 5A06 aluminium alloy as a function of temperature.

Table 2. Physical properties of 5A06 aluminum alloy.

Nomenclature	Symbol	Unit	Value
Solidus temperature	T_S	K	830
Liquidus temperature	T_L	K	940
Specific heat capacity of solid	C_{PS}	$\text{J kg}^{-1} \text{K}^{-1}$	1090
Specific heat capacity of liquid	C_{PL}	$\text{J kg}^{-1} \text{K}^{-1}$	1230
Surface tension coefficient	σ	N m^{-1}	0.914
Temperature coefficient of surface tension	A_δ	$\text{N m}^{-1} \text{K}^{-1}$	-0.00035
Latent heat of fusion	L_m	J kg^{-1}	3.87×10^5
Latent heat of evaporation	L_v	J kg^{-1}	1.08×10^7
Coefficient of thermal expansion	β		1.92×10^{-5}
Convective heat transfer coefficient	h_c	$\text{W K}^{-1} \text{m}^{-2}$	15
Thermal emissivity	ε		0.08
Acceleration due to gravity	g	m s^{-2}	9.8
Ambient temperature	T_{ref}	K	300

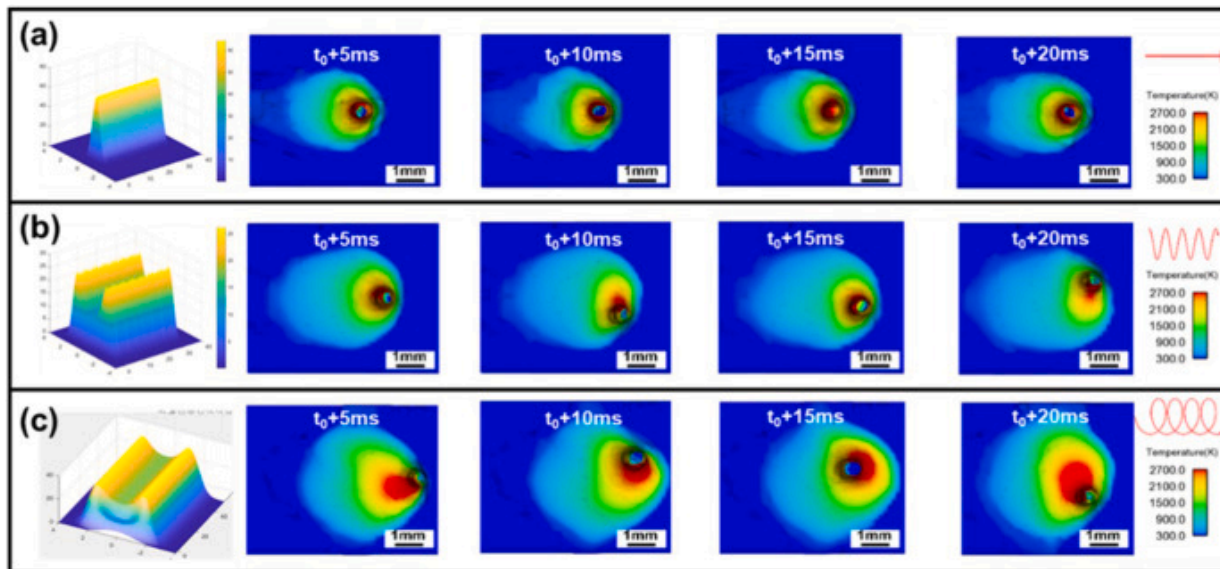
3. Discussion

3.1. Molten pool surface temperature distribution under different laser oscillation modes

Fig. 4 illustrates the molten pool surface temperature distribution under different laser oscillation modes. The underlying mechanisms can be derived from the 3D unsteady heat

conduction theory of moving heat sources. In laser welding, the magnitude and orientation of the thermal gradient are primarily influenced by the energy density distribution and moving speed of the heat source, as well as convection and thermal diffusion within the high-temperature molten pool. For a moving heat source, the temperature field can be described by classical 3D heat conduction [28]:

$$\frac{\partial T}{\partial t} = k \nabla^2 T \quad (25)$$



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Fig. 4. Surface temperature field of molten pool with different oscillation modes: (a) Non-oscillation; (b) Vertical oscillation; (c) Circular oscillation.

In non-oscillating laser welding (Fig. 4a), the heat source energy is concentrated along the keyhole axis, resulting in an elongated molten pool with a distinct tail. The temperature gradient at the front of the heat source can be $2.1 \times 10^6 \text{ K/m}$, which is significantly higher than that in the diffusion zone at the rear. Such a steep gradient intensifies the

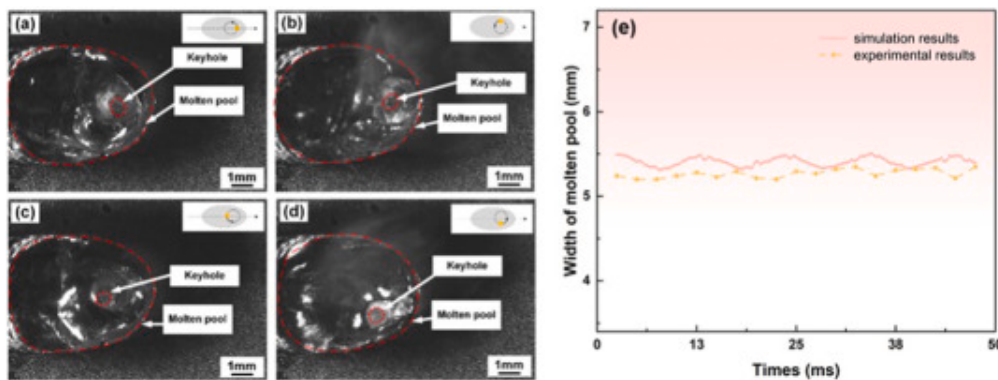
evaporation-induced recoil pressure on the front wall of the keyhole and makes the rear part of the molten pool exposed to continuous cooling. As a result, the susceptibility of the keyhole to contraction and collapse increases, leading to an increased probability of pore formation. In contrast, vertical oscillation (Fig. 4b) distributes the effective heat input over a broader area via lateral, periodic modulation of the heat source position. Consequently, the heat source term $q(r,t)$ in Eq. (1) is transformed from a single-peak distribution into a quasi-periodic one, significantly reducing the temperature gradient to approximately 7.2×10^5 K/m. Additionally, the improved spatial uniformity of heat input enhances the Marangoni circulation in the molten pool, so that high-temperature metal is continuously transported to the pool rear. As a result, the thermal-flow coupling stability around the keyhole is improved. This is consistent with previous studies, wherein laser oscillation can effectively mitigate local thermal concentration, suppress severe keyhole oscillations, and inhibit pore formation [29].

According to Fig. 4c, circular oscillation further expands the range of energy distribution, resulting in a continuous closed time-average trajectory of the heat source. In this case, localized heat generates a quasi-uniform heating ring along the circular region instead of being concentrated at a single center. As a result, a stronger rotational convection is observed in the molten pool. The high-temperature region expands in a ring-like manner, and the molten pool approaches an isothermal state. Its temperature gradient is further reduced to approximately 5.8×10^5 K/m. As the cooling rate at the tail decreases, the residence and buoyancy durations of bubbles in the molten pool are significantly prolonged, facilitating their escape from the liquid surface. Additionally, circular oscillation enhances metal reheating at the tail of the molten pool, thereby decelerating solidification at local solid-liquid interfaces. In this way, periodic contraction and collapse at the rear end of the keyhole are effectively suppressed.

Circular oscillation provides a higher degree of spatial symmetry in energy deposition compared with vertical oscillation. This symmetric distribution promotes the formation of stable horizontal vortex rings within the molten pool, which enhances heat input

homogenization and improves melt pool flow stability. As a result, circular oscillation exhibits superior thermal uniformity and dynamic stability relative to vertical oscillation.

The molten pool width on the upper surface was determined by simulations and high-speed imaging. As shown in Fig. 5, the simulated molten pool size was slightly larger than the experimental one, while the deviation did not exceed 6.1%. The simulated molten pool width exhibited high stability and pronounced periodicity, demonstrating positive effects of circular oscillation laser on molten pool stability. In this way, the proposed model was validated.



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Fig. 5. Comparison of the width of molten pool obtained from experimental and simulation results: (a-d) High-speed photography of the molten pool; (e) Molten pool width statistics.

3.2. Keyhole stability under different laser oscillation modes

The formation and maintenance of the keyhole in laser welding is a typical process involving the coupling of multiple physical fields. It is controlled by evaporation-induced recoil pressure generated by surface vaporization, static pressure of molten metal, surface tension, the Marangoni effect, and inertial flow in the molten pool. Upon interaction of a high-energy-density laser with the material surface, the metal rapidly vaporizes and

ejects outward. The recoil pressure can be described by:

$$P_r + P_v \approx P_h + P_\gamma + P_i \quad (26)$$

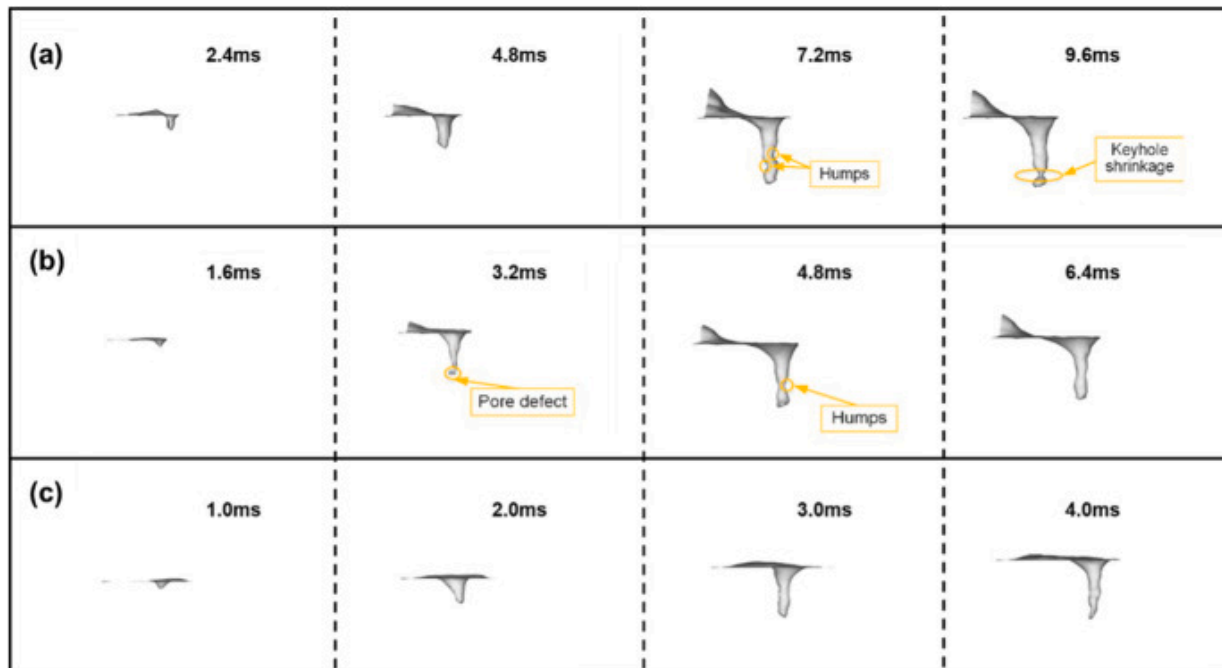
where P_r refers to the evaporation-induced recoil pressure; P_0 refers to the atmospheric pressure; L_v refers to the energy required by the metal transformation of liquid to gas; T_s refers to the surface temperature of the metal; T_b refers to the boiling point of the metal; P_v refers to the internal pressure of the vapor cavity; P_h refers to the static pressure of the liquid column; P_γ refers to the surface tension pressure; P_i refers to the inertial force of the liquid.

If the evaporation-induced recoil pressure exceeds its static pressure, surface tension and inertia, the molten metal is pushed aside and a keyhole is generated. Overall, the keyhole stability can be described by the mechanical equilibrium of the forces mentioned above. Once the evaporation-induced recoil pressure is locally intensified due to temperature fluctuation, wall deformations such as bulging, constriction, and closure are observed, triggering typical keyhole oscillations that directly affect molten pool stability and pore formation.

In non-oscillating laser welding (Fig. 6a), energy is concentrated at a fixed location, causing continuous irradiation on the front wall of the keyhole by a high-power laser. As a result, the evaporation-induced recoil pressure is continuously intensified, resulting in rapid keyhole deepening. In this case, the strong temperature gradient induces significant Marangoni convection, whose surface driving force can be expressed as [30]:

$$v_M \propto \frac{\partial \gamma}{\partial T} \nabla T \quad (27)$$

where $\frac{\partial \gamma}{\partial T}$ refers to the derivative of surface tension on the temperature, and ∇T refers to the temperature gradient.



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Fig. 6. Keyhole formation process during laser welding with different oscillation patterns: (a) Non-oscillation; (b) Vertical oscillation; (c) Circular oscillation.

For aluminum alloys, $\frac{\partial \gamma}{\partial T} < 0$, so that the molten metal moves from cooler regions toward regions of higher temperature along the surface tension gradient, local wall protrusion. It should be noted that the surface tension and viscosity of the molten metal vary with temperature above the liquidus: as temperature increases, surface tension decreases while viscosity is reduced. This temperature-dependent behavior enhances Marangoni convection, strengthens the molten pool flow, and contributes to a more uniform temperature distribution, which helps smooth the pool surface and suppress the formation of unstable protrusions and bottom cavities. Upon superimposition of the vapor jet disturbance with the flow field distribution, this protrusion is readily exposed to instability and collapses inward, generating a bulge-contraction cyclic structure. This

behavior is a key source of keyhole oscillation in non-oscillating laser welding. As a result, enclosed cavities tend to be developed at the keyhole bottom, which serves as the origin of pores.

According to Fig. 6b, the vertical oscillation laser periodically changes the working site of the laser, continuously redistributing energy between the front and rear keyhole walls, thereby significantly reducing sustained local high-energy input. The presence of oscillation reduces the heating time of the laser on a single area, effectively lowering the time-averaged evaporation-induced recoil pressure:

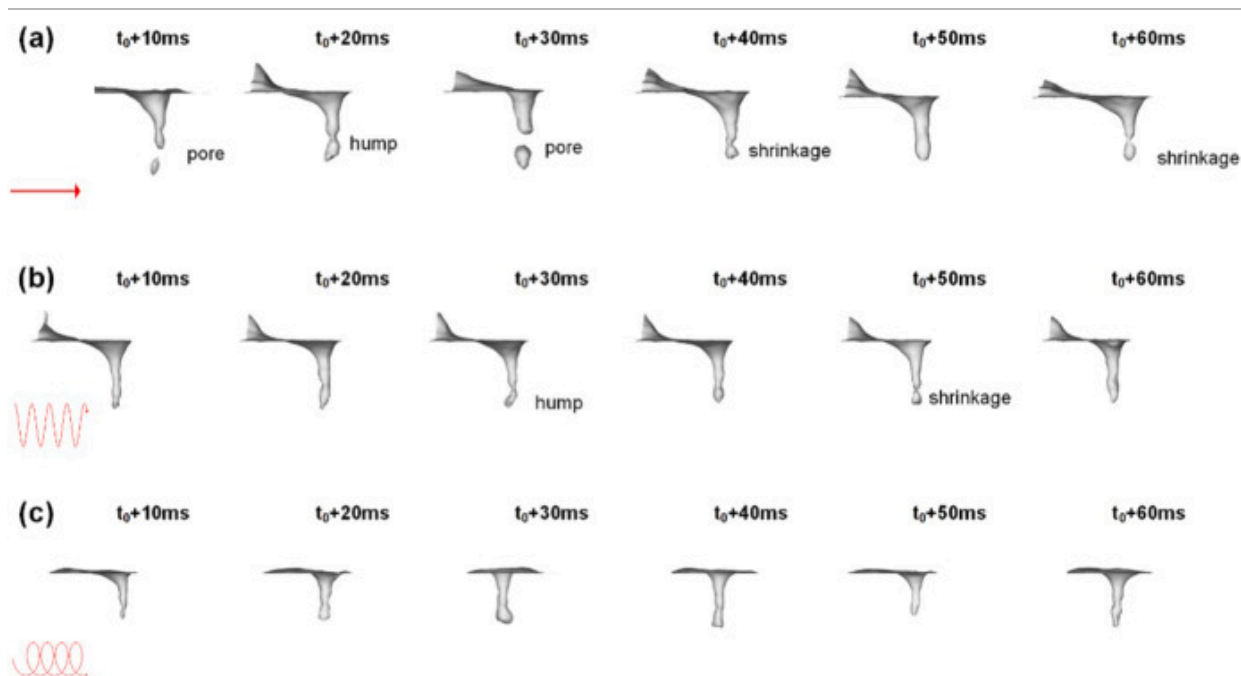
$$P_r = \frac{1}{T} \int_0^T P_r(t) dt \quad (28)$$

Meanwhile, the reduction of temperature gradient leads to a smoother Marangoni-driven flow. As oscillation continues, the density of protrusions on the keyhole wall decreases, the deformation amplitude diminishes, and the geometric contour becomes smoother and more stable. Although localized closure and small cavities may persist during the establishment phase, the overall instability is significantly lower than that in non-oscillating laser welding.

The keyhole establishment process corresponding to circular oscillation is shown in Fig. 6c. As indicated, a keyhole can be developed within 2.0 ms. Its circular trajectory ensures that energy is distributed in a quasi-uniform manner around the keyhole, resulting in a nearly axisymmetric spatial distribution of recoil pressure. This prevents excessive evaporation on the front wall due to localized overheating. Meanwhile, the circular energy input induces a ring-shaped rotating convection within the molten pool, so that the flow lines approach a vortex ring structure. This significantly smooths out the temperature gradient and surface tension gradient, making it difficult for localized peaks to develop into collapse-triggering unstable protrusions. Also, circular oscillation periodically heats the rear wall of the keyhole to decelerate its cooling and effectively prevent collapse. Li et al. [10] also demonstrated that circular oscillation can suppress collapse and inhibit the formation of undercuts. Simulations indicate that the keyhole

wall exhibits negligible deformation, and bottom cavities are unlikely to form. As a result, vapor escape is easier, significantly reducing the tendency for pore formation.

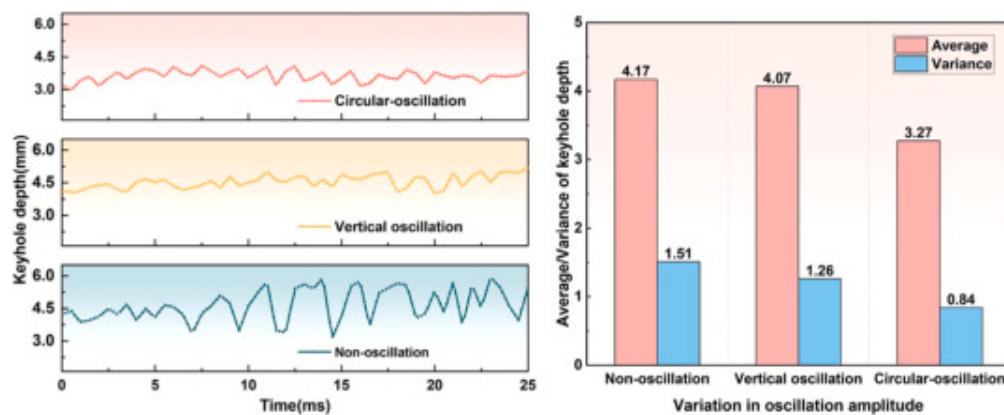
Fig. 7 shows the temporal response characteristics of various keyhole depths as influenced by laser oscillation mode, while Fig. 8 presents the depth ranges and corresponding statistical fluctuation amplitudes. As indicated, the average depths in non-oscillating laser welding, vertical oscillation laser welding, and circular oscillation laser welding were 4.17, 4.07 and 3.27 mm, respectively, with corresponding variances of 1.51, 1.26 and 0.84 mm^2 , respectively. In other words, the average depth converges as the oscillation intensifies. In non-oscillating laser welding, the keyhole depth fluctuates significantly within the range of 3.2–5.9 mm, and the depth curve exhibits wide, irregular fluctuations, indicating frequent and alternating dominance of evaporation-induced recoil pressure and metal backfilling, resulting in repeated collapse and reconstruction of the local wall and highly random flow field disturbances. In vertical oscillation laser welding, the depth range narrows to 3.8–5.2 mm, with noticeably reduced fluctuations. Periodic scanning stabilizes vertical recirculation, facilitating uniform distribution of heat and pressure on a large scale.



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Fig. 7. Dynamic characteristics of keyhole occurred under different oscillation modes: (a) Non-oscillation; (b) Vertical oscillation; (c) Circular oscillation.



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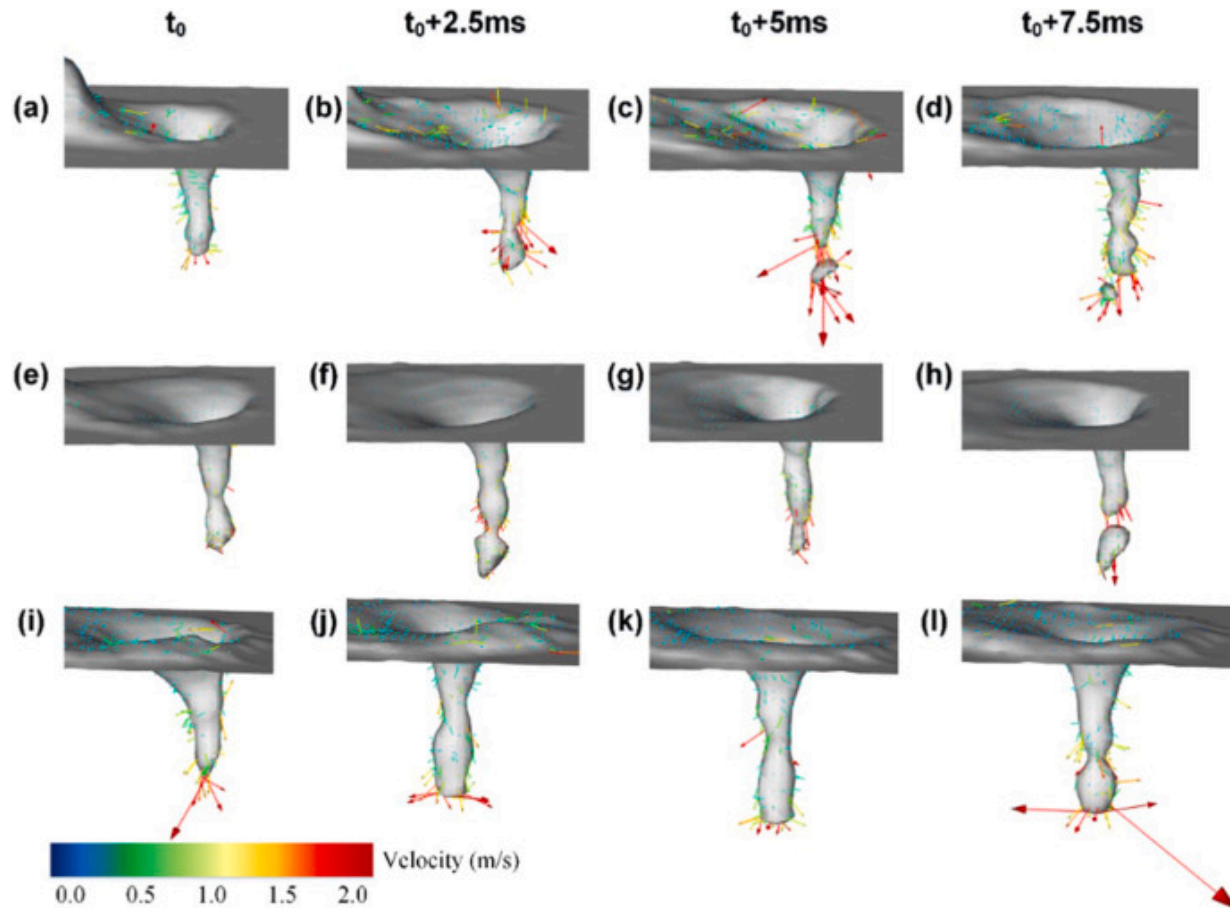
Fig. 8. Influence of oscillation mode on the depth variation of keyhole: (a) Keyhole depth statistics; (b) Keyhole depth mean and variance.

With the circular oscillation pattern, the keyhole depth was confined to a narrow range of 3.0–4.0 mm. The average depth decreased to 3.27 mm, accompanied by a reduced variance of 0.84, indicating the lowest level of fluctuation among the conditions studied. This pattern promoted a more uniform in-plane distribution of laser energy and induced stable horizontal circulatory flow at the base of the molten pool, thereby significantly alleviating uneven stress distribution along the keyhole wall. Under these conditions, simulations show that pronounced collapse is effectively suppressed. The observed depth variations primarily reflect the periodic perturbations introduced by the oscillation trajectory, rather than random flow-field instabilities. Overall, the application of oscillation transforms transient keyhole depth fluctuations from broad and disordered into narrow and regular. The concurrent reduction in mean depth and continued contraction of variance demonstrate that oscillation substantially enhances keyhole stability by restructuring the internal flow dynamics of the molten pool.

3.3. Flow behaviors of the molten pool under different laser oscillation modes

3.3.1. Flow behaviors of keyhole wall

In oscillating laser welding, the keyhole wall flow, the internal circulation of the molten pool and bubble behaviors are determined by strong coupling of the pressure field, temperature field and free surface deformation [31]. [Fig. 9](#) illustrates the 3D flow structure of the keyhole wall and its formation mechanisms under different oscillation modes.



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Fig. 9. The keyhole wall flow behaviors under different oscillation modes: (a-d) Non-oscillation; (e-h) Vertical oscillation; (i-l) Circular oscillation.

In non-oscillating laser welding (Fig. 9a-d), local protrusions on the keyhole wall are exposed to drastic intensification of pressure gradient and surface tension gradient caused by increasing curvature. Therefore, the evaporation-induced recoil pressure and the Marangoni shear force drive the liquid to flow downward rapidly along the wall. High-curvature zone at the keyhole bottom serves as a local pressure sink, further guiding the

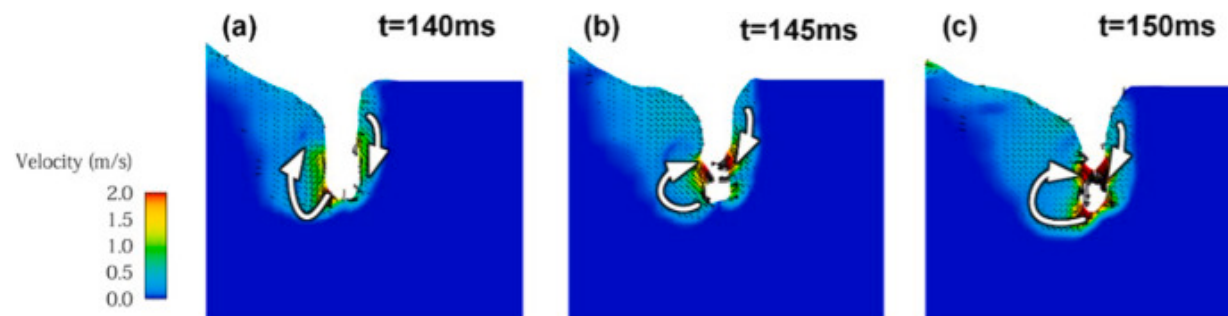
melt downward. In cases of transient closure of the keyhole, the lower bubble interface is dragged and detaches from the main keyhole cavity. If their velocity exceeds the shrinkage rate of the keyhole, bubbles may be trapped in the molten pool and subsequently generate pore defects during solidification.

In vertical oscillation laser welding (Fig. 9e-h), the periodic scanning of the laser beam reduces the peak temperature of the keyhole wall, thereby reducing the amplitude of the Marangoni force. As a result, the density of wall protrusions significantly decreases, and the pressure gradient becomes more uniform. Additionally, the frequency of keyhole collapses decreases drastically, and the interfacial flow rate of bubbles, if any, is substantially reduced, indicating a weakened dragging of the bubbles. Consequently, bubbles are more likely to escape. Overall, vertical oscillation effectively suppresses unstable disturbances caused by the flow field-wall-pressure coupling, significantly reducing the risk of pore formation.

In circular oscillation laser welding (Fig. 9i-l), the continuous and uniform energy input further smooths the wall temperature field, markedly dampening pressure field fluctuations and suppressing excessive deformation of the free surface. As the laser spot moves along a circular trajectory, the keyhole wall morphology is no longer featured by sharp corners and high curvature, resulting in stabilized evaporation-induced recoil pressure and increased symmetry of the flow field. High-velocity flow regions are primarily concentrated at the keyhole bottom, while the average flow velocity on the free surface is significantly reduced compared to those in the previous two cases, indicating that circular oscillation effectively suppresses the nonlinear flow field - keyhole geometry - pressure field coupling instability, reducing the likelihood of keyhole collapse and pore formation. Overall, circular oscillation leads to the highest keyhole stability and the most significant pore suppression. Overall, circular oscillation exhibits optimized keyhole stability and pore formation capability.

3.3.2. Flow behaviors of the molten pool at keyhole rear

Fig. 10 illustrates the flow field at the central longitudinal cross-section of the molten pool in non-oscillating laser welding. Driven by the evaporation-induced recoil pressure, the melt behind the keyhole flows backward and upward from the bottom, resulting in a noticeable backflow vortex (**Fig. 10**). The flow velocity at the keyhole bottom tends to be high, which can be attributed to the small distance between the keyhole bottom and the solidified material beneath. Also, the recoil pressure enhances fluid acceleration in this narrow space. This is consistent with previous studies [32]. Likewise, local protrusions on the keyhole wall absorb more laser energy due to irregular geometry, leading to increased local temperature gradient and surface tension gradient. This accelerates the melt toward the protrusion zone, resulting in a clockwise vortex. The maximum flow velocity (~ 3.2 m/s) in the molten pool was observed near the front wall of the keyhole. Hence, it can be deduced that the flow field in non-oscillating laser welding is featured by a distinct upward-return flow and local convergence, leading to uneven stress on the wall and increased curvature. Consequently, keyhole instability or even collapse is triggered.



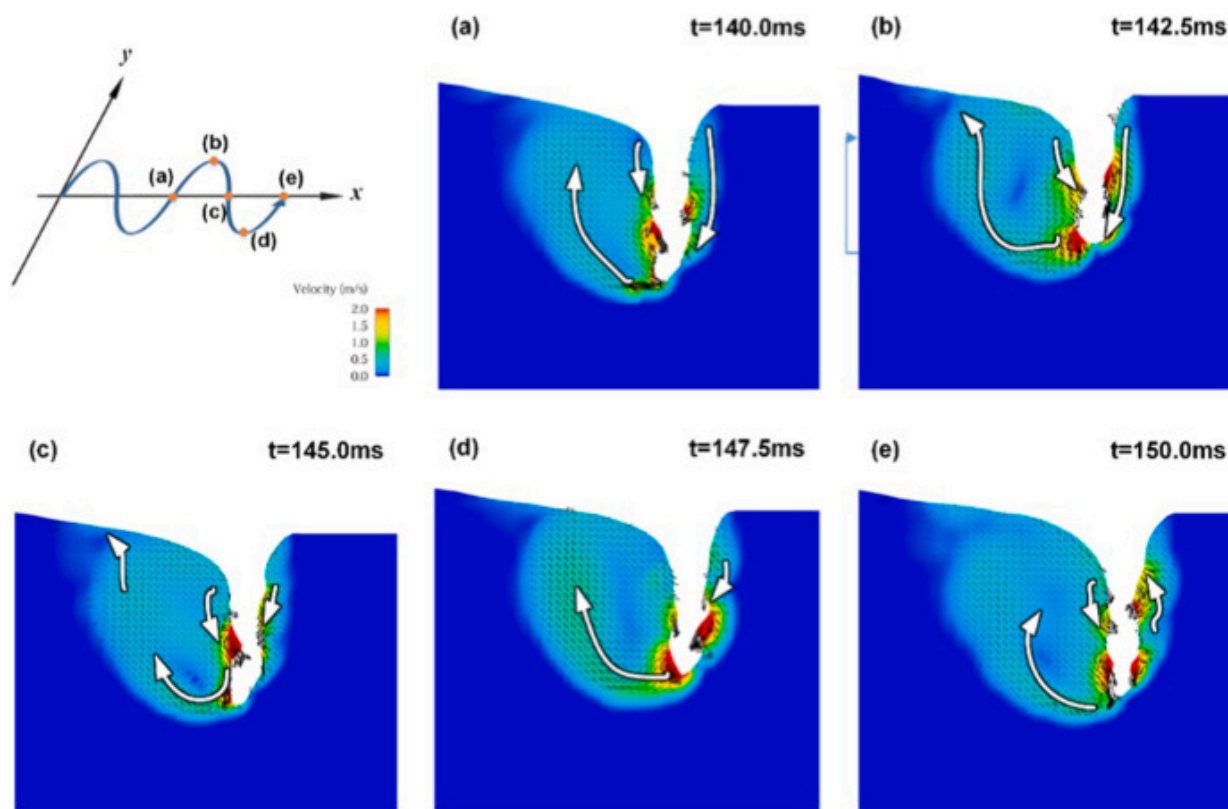
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Fig. 10. Molten pool flow patterns during non-oscillating laser welding.

In vertical oscillation laser welding (**Fig. 11**), the flow behaviors of the longitudinal cross-section of the molten pool show similar characteristics at different moments. The overall flow field exhibits a downward-backward flow behind the keyhole, which is consistent with the non-oscillating laser welding case. Nevertheless, local fluid disturbances are

different. Particularly, a counterclockwise vortex is present near the wall protrusion, as shown in Fig. 11c. This can be attributed to the velocity differences caused by uneven local energy absorption. This counter vortex is unfavorable for the downward expansion of the keyhole and may induce local collapse and promote bubble formation. Despite this, vertical oscillation effectively reduces the flow velocity of the molten pool and increases the uniformity of velocity distribution [33]. It leads to significantly reduced intensity of the upward-return flow, so that the maximum flow velocity drops to ~ 1.9 m/s. The enhanced uniformity of the flow field implies more stable stress on the keyhole wall, lower density of local high-curvature protrusions, and significantly reduced collapse risks.

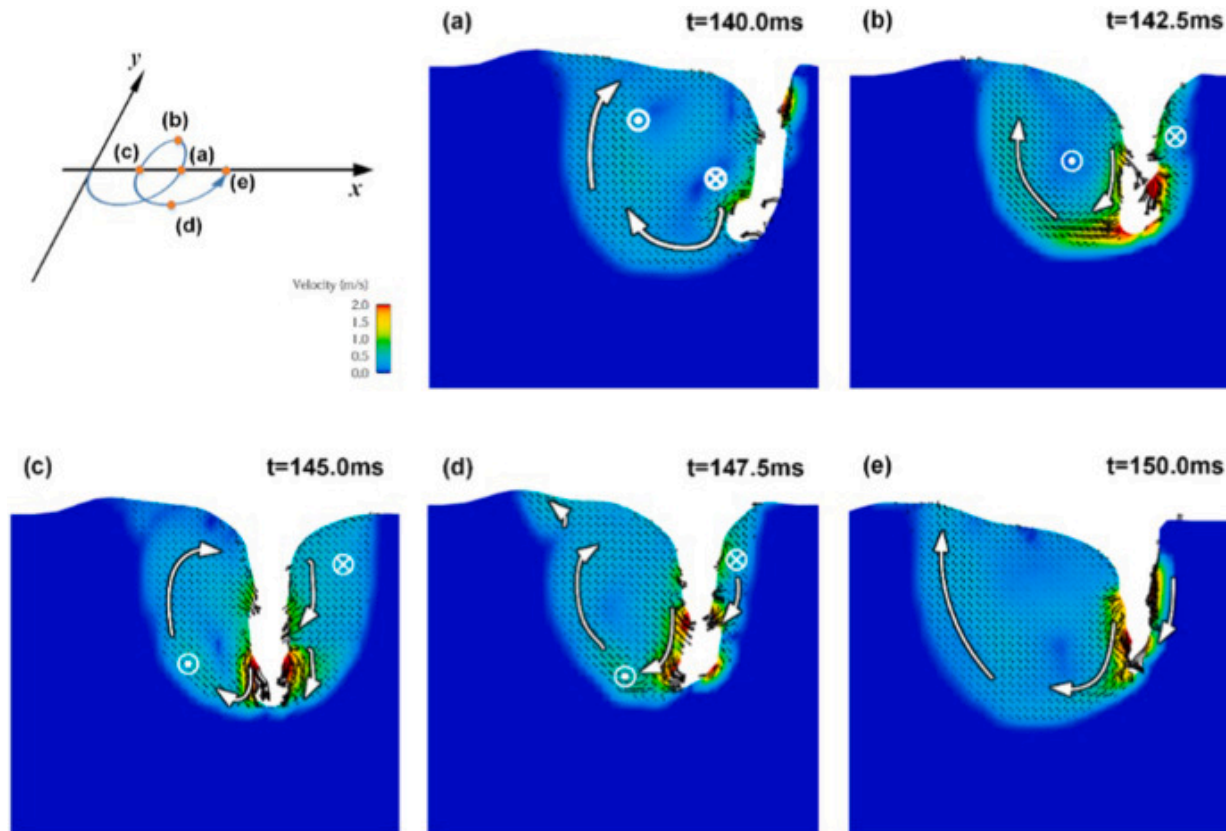


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Fig. 11. Molten pool flow patterns during vertical oscillation welding.

Fig. 12 shows the flow behaviors of the molten pool under circular oscillation laser welding. Since the laser beam moves backward relative to the molten pool at some moments during oscillation, its position change significantly affects the flow structure inside the keyhole. In circular oscillation laser welding, the laser beam triggers continuously stirring of the molten pool, generating stable and strong horizontal vortices inside the molten pool, whose direction consistently aligns with the beam motion. Once the beam is at the front of the oscillation trajectory (Fig. 12a), the keyhole wall is located at the front of the molten pool. Horizontal vortices concentrate behind the keyhole, resulting in high flow stability in the molten pool, uniform velocity distribution, and negligible wall protrusions. As the beam moves to the designated position (Fig. 12b), the vortex shifts forward and bypasses the keyhole, causing minor protrusions on the wall. Nevertheless, no unfavorable counterclockwise vortices are generated, and no collapses are triggered. When the keyhole is located in the center of the molten pool (Fig. 12c), the melt flows from the front of the keyhole toward its bottom or from the rear of the keyhole toward the tail of the molten pool. Despite a slight increase in the local flow velocity on the free surface, no vortices hindering the keyhole opening are generated. As the beam moves further (Fig. 12d-e), the horizontal vortices gradually degrade, and the keyhole wall becomes smoother. In this process, the average flow velocity in the molten pool decreases further, with a maximum flow velocity of ~ 0.8 m/s, which is significantly lower than those in the other two cases. Overall, circular oscillation significantly reduces local pressure gradients and temperature gradients via continuous energy smoothing and beam traction, inhibiting the formation of unfavorable vortices. Consequently, the keyhole stability is significantly improved.



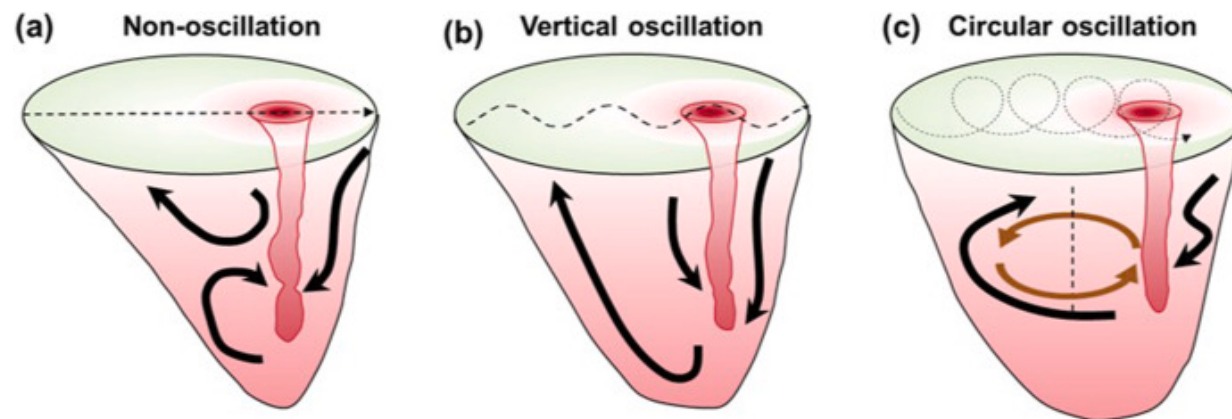
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Fig. 12. Melt flow patterns for laser welding with circular oscillation.

According to Fig. 13, the oscillation mode has a substantial influence on the flow structure in the molten pool. In non-oscillating laser welding, the pronounced reverse-flow vortex at the rear of the keyhole causes high-speed upward return flow of the lower melt, impacting the keyhole wall. Under uneven stress conditions, this easily induces wall deformation, protrusion, and even collapse. Meanwhile, the upper vortex rapidly transports the melt along the surface to the tail of the molten pool, thereby hindering the exchange between the upper and lower melt and reducing the probability of bubble

escape. Both vertical oscillation and circular oscillation can effectively suppress this strong vortex, so that the melt near the keyhole wall exhibits a smooth downward flow along the wall. The local impact on the wall caused by vertical oscillation persists, while the intensity is significantly reduced. In contrast, the horizontal vortices generated by circular oscillation can draw the melt near the wall away from the keyhole, fundamentally reducing the impact on the wall. In this way, the keyhole stability can be improved and the risk of bubble generation and retention can be significantly reduced. Additionally, circular oscillation leads to an expanded upper surface area of the molten pool and reduced internal temperature gradient, resulting in increased uniformity of the flow field. This favors the uniform distribution of alloy elements during solidification, as well as the inhibition of pore defects.



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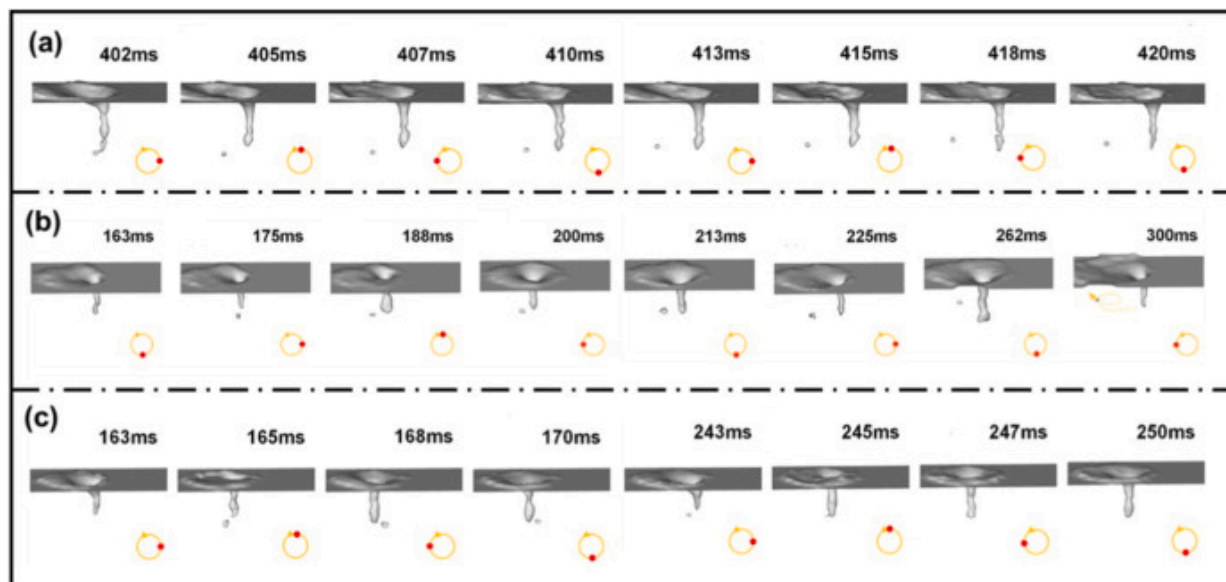
Fig. 13. Comparison of molten pool flow characteristics under different laser oscillation modes: (a) Non-oscillation; (b) Vertical oscillation; (c) Circular oscillation.

3.4. Mechanism of pore generation

According to the coupling mechanism of the flow field and keyhole dynamics, the oscillation parameters of the laser beam (A =amplitude, f =frequency) profoundly affect

the thermal cycle and flow structure of the molten pool by altering the spatiotemporal characteristics of energy deposition, thereby determining the keyhole stability and the mechanism of bubble defect formation. In laser welding of aluminum alloys, high thermal conductivity and active surface tension effects of aluminum alloys make the molten pool behaviors particularly sensitive to the energy input mode.

At low oscillation amplitudes (Fig. 14a), the periodic deviation of the beam is substantially smaller than the characteristic size of the keyhole (depth and opening diameter). From the thermophysical perspective, the heat-affected zones covered by the beam trajectory highly overlap, and its time-averaged spatial distribution of energy is essentially consistent with the non-oscillation laser welding case. Therefore, the periodic motion of the laser beam fails to trigger effective fluid mechanical agitation in the molten pool and fails to significantly alter the flow field structure dominated by the Marangoni convection driven by steep temperature gradients and intense metal vapor jets [25]. The keyhole wall exhibits random protrusions, which are a direct result of the dynamic imbalance between the local evaporation-induced recoil pressure and the surface tension varying drastically with temperature (consistent with the characteristics of the Marangoni effect of the aluminum alloy molten pool). When the local wall curvature exceeds a critical level due to overheating, the surface tension can no longer balance the internal pressure, causing wall contraction. This triggers transient local closure of the keyhole, trapping vapor and generating bubbles. Due to the limited motion range of the beam, the induced disturbance in the flow field can hardly alter the balance of drag and buoyancy forces on the bubbles or create effective escape channels for the bubbles. Eventually, bubbles are easily captured by the rapidly advancing solidification front of the molten pool, resulting in the formation of pore defects. At this stage, the keyhole behaviors are still dominated by their inherent thermal-mechanical instability, and low-amplitude oscillation does not provide sufficient intervention.



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Fig. 14. The formation process of pores under different oscillation modes: (a) Non-oscillation; (b) Vertical oscillation; (c) Circular oscillation.

With increasing oscillation amplitude (Fig. 14b), the spatial scanning range of the beam exceeds the transient size of the keyhole, thereby enabling periodic disturbance and reconstruction of the molten pool flow field. Consequently, the keyhole morphology and the surrounding flow exhibit regular variations: at the endpoints (front and rear parts) of the beam trajectory where the beam moves rapidly, the energy deposition duration is short, the keyhole length is short, and fluctuations in wall evaporation pressure are reduced due to relatively uniform heat input, resulting in reduced wall roughness; on the sides of the trajectory where the beam moves slowly, energy accumulates, the keyhole extends, and the instability of evaporation pressure is intensified by local overheating, leading to the reappearance of wall protrusions. Nevertheless, the characteristic motion time of the keyhole does not match the response time of the fluids in the molten pool under low oscillation frequencies, and its average velocity remains low. Although the

enhanced stirring effect disrupts the original quasi-steady-state flow field, it extends the retention time and travel distance of bubbles formed in the molten metal. Due to the limited overall migration speed of the keyhole, the solidification interface at the front of the molten pool may have already advanced beyond the bubbles when the keyhole moves to the trailing edge, preventing them from being recaptured or effectively expelled by the keyhole flow field. Therefore, such a parameter combination may alter the mechanism of pore generation from “direct keyhole collapse and encapsulation” to “bubble capture by the solidification interface after migration and entrapment in a complex flow field”, meaning the risk of defects persists.

When high amplitude and high frequency are employed (Fig. 14c), the rapid, large-scale scanning of the laser fundamentally regulates the molten pool. Under high-frequency oscillation, the laser energy covers a broad area within an extremely short period, which is equivalent to averaging the heat input both temporally and spatially. This significantly reduces the transient fluctuating amplitude of evaporation-driven recoil pressure exerted on the local keyhole wall, thereby effectively suppressing the formation of irregular wall wrinkles and severe contraction instabilities. As a result, the keyhole morphology becomes smooth and stable [34]. A more decisive mechanism originates from the synergy of the dynamic flow field induced by high-amplitude oscillations and the motion trajectory of the keyhole. Specifically, the keyhole is essentially a high-temperature, low-pressure cavity that undergoes large-scale, high-speed motion along with the beam. When bubbles are generated at the keyhole bottom due to transient fluctuations and are left behind at the original location, the wall or wake of the rapidly moving keyhole cavity can pass through the area where the bubbles are located again in subsequent oscillation cycles. In the molten pool of aluminum alloys, the strong convection induced by the large-scale beam motion, combined with the pressure gradient generated by the keyhole motion, can drive bubbles toward the keyhole, even re-entraining them into the high-temperature cavity. Under the high heat flow, the recaptured bubbles may collapse, dissolve or vanish when ejected with the metal vapor. This “dynamic tracking and

elimination” process is the core physical mechanism by which high-amplitude, high-frequency oscillations significantly reduce welding porosity in aluminum alloys.

4. Conclusion

This study analyzes molten pool flow and pore formation during oscillating laser welding of aluminum alloy plates, considering different oscillation patterns. The results are directly applicable to 5A06 aluminum alloy. However, mechanisms such as Marangoni convection and thermo-physical effects are general and may extend to other aluminum alloys or plate thicknesses, though variations in composition or thickness may require parameter adjustment.

- (1) Vertical and circular oscillations can reduce the surface temperature gradient of the molten pool by expanding the area of the energy-affected zone, thereby effectively alleviating the local heat concentration. The annular energy distribution in the case of circular oscillation results in the most convergent fluctuation of the keyhole depth, achieving a transition from random instability to periodic controllable responses.
- (2) In non-oscillating laser welding, the strong upward-return vortex is the main cause of interface deformation and collapse. Vertical oscillation can reduce the vortex intensity, while circular oscillation generates horizontal vortex rings consistent with beam motion, thereby significantly reducing impact on the wall and the flow velocity of the molten pool. In this way, a stable interfacial environment is established from a fluid mechanics perspective.
- (3) Low-amplitude or low-frequency oscillations can hardly suppress bubble formation, while medium parameters may prolong the retention of bubbles. High-amplitude, high-frequency oscillation can fundamentally suppress keyhole contraction-induced bubble generation and achieve recapturing

and dissolving of existing bubbles based on the periodic motion of the keyhole, ultimately achieving the optimized porosity suppression effect.

CRediT authorship contribution statement

Chen Liu: Formal analysis, Methodology, Writing – original draft. **Yuanze Ma:** Software. **Junzhuo Guo:** Software. **Jianfeng Gong:** Conceptualization, Formal analysis, Writing – review & editing. **Haoyue Li:** Conceptualization. **Yingke Wang:** Writing – review & editing. **Liqun Li:** Supervision. **Changyun Li:** Writing – review & editing. **Hongbo Xia:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] H. Liu, H. Zhao, Y. Zhao, Y. Peng, M. Xu, X. Chen
Interfacial bonding mechanisms in laser welding of 2024 Al alloy and continuous CFR-PEEK via adjustable ring-mode laser beam
J. Manuf. Process., 153 (2025), pp. 757-773, [10.1016/j.jmapro.2025.09.020](#) ↗

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [2] B. Stojanovic, M. Bukvic, I. Epler
Application of aluminum and aluminum alloys in engineering
Appl. Eng. Lett. J. Eng. Appl. Sci., 3 (2018), pp. 52-62, [10.18485/aeletters.2018.3.2.2 ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [3] Z. Wang, J. Song, X. Li, C. Liu, M. Liu, L. Li, X. Xu, Y. Wang, J. Ni, B. Song
Advanced current collector design for high energy density anode-free sodium-ion batteries
Electron, 3 (2025), Article e70013, [10.1002/elt2.70013 ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [4] C. Liu, Y. Ma, S. Zhang, C. Li, L. Li, J. Wang, H. Li, C. Tan, H. Xia, P. He
Influence of laser power on weld formation, microstructure, and mechanical properties of Q235B steel joined by laser-CMT hybrid welding process
Int. J. Pres. Ves. Pip., 219 (2026), Article 105688, [10.1016/j.ijpvp.2025.105688 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [5] C. Liu, Y. Ma, P. Guo, H. Li, Y. Wang, J. Gong, L. Li, C. Li, H. Xia
Effects of oscillation amplitude and frequency in narrow-gap laser wire-filling welding of Al alloy 5A06 on the melt pool and the mechanism of pore inhibition
Int. J. Therm. Sci., 221 (2026), Article 110494, [10.1016/j.ijthermalsci.2025.110494 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [6] J. Enz, M. Kumar, S. Riekehr, V. Ventzke, N. Huber, N. Kashaev
Mechanical properties of laser beam welded similar and dissimilar aluminum alloys
J. Manuf. Process., 29 (2017), pp. 272-280, [10.1016/j.jmapro.2017.07.030 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [7] A.C. Popescu, C. Delval, S. Shadman, M. Leparoux
Investigation and in situ removal of spatter generated during laser ablation of aluminium composites
Appl. Surf. Sci., 378 (2016), pp. 102-113, [10.1016/j.apsusc.2016.03.031](https://doi.org/10.1016/j.apsusc.2016.03.031) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [8] J. Lin, Y. Chen, D. Ma, L. Xing, P. He, W. Yang, T. Lin
Achieving nearly-homogeneous joint of ZrC-SiC composite via liquid-film assisted transformation strategy enabled by pulsed current
J. Mater. Sci. Technol., 243 (2026), pp. 192-205, [10.1016/j.jmst.2025.03.091](https://doi.org/10.1016/j.jmst.2025.03.091) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [9] R. Zhang, D. Nishimoto, N. Ma, K. Narasaki, Q. Wang, T. Suga, S. Tsuda, T. Tabuchi, S. Shimada
Asymmetric molten zone and hybrid heat source modeling in laser welding carbon steel and cast iron with nickel alloy wires
J. Manuf. Process., 142 (2025), pp. 177-190, [10.1016/j.jmapro.2025.03.074](https://doi.org/10.1016/j.jmapro.2025.03.074) ↗
 [View PDF](#) [View article](#) [Google Scholar](#) ↗
- [10] Z. Li, F. Li, C. Shen, Z. Li, Y. Chen, X. Hua
Undercut suppression via time-dependent laser power modulation strategy during the circular oscillating laser welding process
J. Manuf. Process., 152 (2025), pp. 1090-1110, [10.1016/j.jmapro.2025.08.074](https://doi.org/10.1016/j.jmapro.2025.08.074) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [11] C. Cai, Y. He, E. Wang, Z. Xian, H. Chen
Ductility enhancement mechanism of laser welded magnesium alloy joints via synergistic effect of Gd powder addition and oscillating laser
J. Magnesium Alloys (2025), [10.1016/j.jma.2025.08.010](https://doi.org/10.1016/j.jma.2025.08.010) ↗
S2213956725002774
[Google Scholar](#) ↗

- [12] M. Wu, Z. Luo, Y. Li, L. Liu, S. Ao
Effect of oscillation modes on weld formation and pores of laser welding in the horizontal position
Opt. Laser Technol., 158 (2023), Article 108801, [10.1016/j.optlastec.2022.108801](https://doi.org/10.1016/j.optlastec.2022.108801) ↗
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [13] F. Fetzner, M. Sommer, R. Weber, J.-P. Weberpals, T. Graf
Reduction of pores by means of laser beam oscillation during remote welding of AlMgSi
Opt Laser. Eng., 108 (2018), pp. 68-77, [10.1016/j.optlaseng.2018.04.012](https://doi.org/10.1016/j.optlaseng.2018.04.012) ↗
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [14] P. Xia, C. Wang, G. Mi, M. Zhang, L. Xiong, X. Zhang, C. Zhai, X. Feng, Y. Hu
Numerical simulation of molten pool flow behavior and keyhole evolution behavior in dual-laser beam oscillating bilateral synchronous welding of T-joints
Int. J. Heat Mass Tran., 209 (2023), Article 124114, [10.1016/j.ijheatmasstransfer.2023.124114](https://doi.org/10.1016/j.ijheatmasstransfer.2023.124114) ↗
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [15] W.-I. Cho, P. Woizeschke
Analysis of molten pool dynamics in laser welding with beam oscillation and filler wire feeding
Int. J. Heat Mass Tran., 164 (2021), Article 120623, [10.1016/j.ijheatmasstransfer.2020.120623](https://doi.org/10.1016/j.ijheatmasstransfer.2020.120623) ↗
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [16] H. Deng, Y. Li, Z. Peng, Y. Huang, W. Wei, Y. Long
A molten pool and keyhole dynamics study in near IR-blue hybrid laser welding of AZ31B magnesium alloy: a dual-mode synergy strategy for defect suppression in reflective low-melting materials
J. Mater. Process. Technol., 339 (2025), Article 118814, [10.1016/j.jmatprotec.2025.118814](https://doi.org/10.1016/j.jmatprotec.2025.118814) ↗

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [17] H. Kim, K. Nam, Y. Kim, H. Ki
Analysis of laser-beam absorptance and keyhole behavior during laser keyhole welding of aluminum alloy using a deep-learning-based monitoring system

J. Manuf. Process., 80 (2022), pp. 75-86, [10.1016/j.jmapro.2022.05.044 ↗](#)

 [View PDF](#) [View article](#) [Google Scholar ↗](#)

- [18] V.R. Voller, C. Prakash
A fixed grid numerical modeling methodology for convection-diffusion mushy region phase-change problems

Int. J. Heat Mass Tran., 8 (1987), pp. 1709-1719, [10.1016/0017-9310\(87\)90317-6 ↗](#)

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [19] M. Dal, R. Fabbro
An overview of the state of art in laser welding simulation

Opt. Laser Technol., 78 (2016), pp. 2-14, [10.1016/j.optlastec.2015.09.015 ↗](#)

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [20] S. Pang, X. Chen, J. Zhou, X. Shao, C. Wang
3D transient multiphase model for keyhole, vapor plume, and weld pool dynamics in laser welding including the ambient pressure effect

Opt Laser. Eng., 74 (2015), pp. 47-58, [10.1016/j.optlaseng.2015.05.003 ↗](#)

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [21] B. Huang, X. Chen, S. Pang, R. Hu
A three-dimensional model of coupling dynamics of keyhole and weld pool during electron beam welding

Int. J. Heat Mass Tran., 115 (2017), pp. 159-173, [10.1016/j.ijheatmasstransfer.2017.08.010 ↗](#)

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [22] Z. Gao, P. Jiang, G. Mi, L. Cao, W. Liu
Investigation on the weld bead profile transformation with the keyhole and molten pool dynamic behavior simulation in high power laser welding
Int. J. Heat Mass Tran., 116 (2018), pp. 1304-1313, [10.1016/j.ijheatmasstransfer.2017.09.122](https://doi.org/10.1016/j.ijheatmasstransfer.2017.09.122) ↗
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [23] X. Zhao, J. Chen, Z. Lei, W. Zhang, H. Chen
A study on the flow behavior and bubble evolution of circular oscillating laser welding of SUS301L-HT stainless steel
Int. J. Heat Mass Tran., 202 (2023), Article 123726, [10.1016/j.ijheatmasstransfer.2022.123726](https://doi.org/10.1016/j.ijheatmasstransfer.2022.123726) ↗
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [24] G.A. Taylor, M. Hughes, N. Strusevich, K. Pericleous
Finite volume methods applied to the computational modelling of welding phenomena
Appl. Math. Model., 26 (2002), pp. 311-322, [10.1016/S0307-904X\(01\)00063-4](https://doi.org/10.1016/S0307-904X(01)00063-4) ↗
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [25] R. Zhang, X. Tang, L. Xu, F. Lu, H. Cui
Study of molten pool dynamics and porosity formation mechanism in full penetration fiber laser welding of Al-alloy
Int. J. Heat Mass Tran., 148 (2020), Article 119089, [10.1016/j.ijheatmasstransfer.2019.119089](https://doi.org/10.1016/j.ijheatmasstransfer.2019.119089) ↗
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [26] Y. Ai, P. Jiang, C. Wang, G. Mi, S. Geng
Experimental and numerical analysis of molten pool and keyhole profile during high-power deep-penetration laser welding
Int. J. Heat Mass Tran., 126 (2018), pp. 779-789, [10.1016/j.ijheatmasstransfer.2018.05.031](https://doi.org/10.1016/j.ijheatmasstransfer.2018.05.031) ↗
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [27] Y. Lu, Y. Deng, L. Shi, L. Jiang, M. Gao
Numerical simulation of thermal flow dynamics in oscillating laser welding of aluminum alloy
Opt. Laser Technol., 159 (2023), Article 109003, [10.1016/j.optlastec.2022.109003](https://doi.org/10.1016/j.optlastec.2022.109003) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [28] C.H. Forsberg
Heat conduction equation and boundary conditions
Heat Transf. Princ. Appl., Elsevier (2021), pp. 23-55, [10.1016/B978-0-12-802296-2.00002-0](https://doi.org/10.1016/B978-0-12-802296-2.00002-0) ↗
 [View PDF](#) [View article](#) [Google Scholar](#) ↗
- [29] J. Dakka, H.P. Wang, B. Yang, J. Ma, B. Carlson, J. Oswald
Influence of beam oscillation on keyhole stability and porosity in aluminum laser welding
J. Mater. Process. Technol., 342 (2025), Article 118945, [10.1016/j.jmatprotec.2025.118945](https://doi.org/10.1016/j.jmatprotec.2025.118945) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [30] P. Sahoo, T. Debroy, M.J. McNallan
Surface tension of binary metal—surface active solute systems under conditions relevant to welding metallurgy
Metall. Trans. B, 19 (1988), pp. 483-491, [10.1007/BF02657748](https://doi.org/10.1007/BF02657748) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [31] C. Cai, Y. He, J. Xie, J. Yu, L. Xu, Z. Chen, Z. Li, E. Wang, H. Chen
Porosity suppression mechanism analysis in narrow-gap oscillating laser-MIG hybrid welding of aluminum alloys based on keyhole stability and molten pool flow behavior
J. Mater. Res. Technol., 32 (2024), pp. 502-518, [10.1016/j.jmrt.2024.07.133](https://doi.org/10.1016/j.jmrt.2024.07.133) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [32] Z. Zheng, C. Shao, M. Liu, F. Lu

Formation and migration of bubbles under different laser oscillation paths during laser welding process of medium-thick Al alloy

Int. J. Heat Mass Tran., 239 (2025), Article 126553, [10.1016/j.ijheatmasstransfer.2024.126553](https://doi.org/10.1016/j.ijheatmasstransfer.2024.126553) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [33] J. Shui, H. Liu, Z.-Y. Bi, H.-T. Wang, L.-Y. Xian, W.-D. Wang
Effects of oscillating parameters on weld formation, molten pool morphology, and energy distribution during laser oscillating welding of Ti6Al4V alloy

Opt. Laser Technol., 194 (2026), Article 114447, [10.1016/j.optlastec.2025.114447](https://doi.org/10.1016/j.optlastec.2025.114447) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [34] M. Rupp, K. Schwarzkopf, M. Döring, S. Hayashi, M. Schmidt, C.B. Arnold
Laser driven melt pool resonances through dynamically oscillating energy inputs

J. Manuf. Process., 131 (2024), pp. 1624-1630, [10.1016/j.jmapro.2024.09.100](https://doi.org/10.1016/j.jmapro.2024.09.100) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

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